
ARTIFICIAL LIGHT AT NIGHT AND BAT ACTIVITY AT COLTER BAY

ALONG A DISTANCE GRADIENT

Grand Teton National Park, 2022

Nigel K. Anderson · American Museum of Natural History

Contents

1	Overview	1
2	Primary Model	2
3	Post-hoc: Color x Intensity	5
4	Post-hoc: Intensity within Color	7
5	Post-hoc: Distance	9
6	Community Analysis	12
7	Species-level Results	16
8	Summary	16
9	Species Key	17

OVERVIEW

Study question: does light colour (red vs. white) affect bat activity, and does the answer depend on light intensity or distance from Colter Bay parking lot?

Design: 14 acoustic monitoring stations along a distance gradient (<0.52 km to >2.1 km from Colter Bay parking lot), red vs. white light crossed at five intensities (10, 30, 50, 70, 100%). Survey season: 2022. Eight focal species: Epfu, Laci, Lano, Myev, Mylu, Myvo, Myyu, Myci.

PRIMARY MODEL

Family: nbinom2 (log)

Formula: detections ~ color * intensity + intensity * dist_km + mean_phase +
 pct_nonforest + brightness_dark + jd_c + I(jd_c^2) + (1 | site)

Zero inflation: ~1

Data: data_env

AIC	BIC	logLik	-2*log(L)	df.resid
24256.7	24398.1	-12105.4	24210.7	3434

Random effects:

Conditional model:

Groups	Name	Variance	Std.Dev.
site	(Intercept)	0.3728	0.6106

Number of obs: 3457, groups: site, 14

Dispersion parameter for nbinom2 family (): 0.607

Conditional model:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	2.1920095	0.3044757	7.199	6.05e-13	***
colorW	0.0499199	0.0930811	0.536	0.591747	
intensity30	0.7155529	0.1475594	4.849	1.24e-06	***
intensity50	0.4606876	0.1569073	2.936	0.003324	**
intensity70	0.6229863	0.1098612	5.671	1.42e-08	***
intensity100	0.2331022	0.1330698	1.752	0.079821	.
dist_km	0.1375974	0.0674576	2.040	0.041374	*
mean_phase	0.1992094	0.1170827	1.701	0.088860	.
pct_nonforest	0.5326479	0.8559351	0.622	0.533745	
brightness_dark	-0.0262336	0.0376003	-0.698	0.485367	
jd_c	-0.0073183	0.0017073	-4.287	1.81e-05	***
I(jd_c^2)	-0.0003733	0.0001736	-2.150	0.031522	*
colorW:intensity30	-0.7703147	0.1750938	-4.399	1.09e-05	***
colorW:intensity50	-0.5357506	0.1458849	-3.672	0.000240	***
colorW:intensity70	-0.1550017	0.1365976	-1.135	0.256488	
colorW:intensity100	0.0172641	0.1352882	0.128	0.898457	

```

intensity30:dist_km -0.0501091  0.0318723 -1.572 0.115908
intensity50:dist_km -0.0364428  0.0294086 -1.239 0.215276
intensity70:dist_km -0.1275481  0.0272111 -4.687 2.77e-06 ***
intensity100:dist_km -0.0897028  0.0269009 -3.335 0.000854 ***

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Zero-inflation model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-21.53	805.37	-0.027	0.979

Reference levels: color = R (red), intensity = 10%.

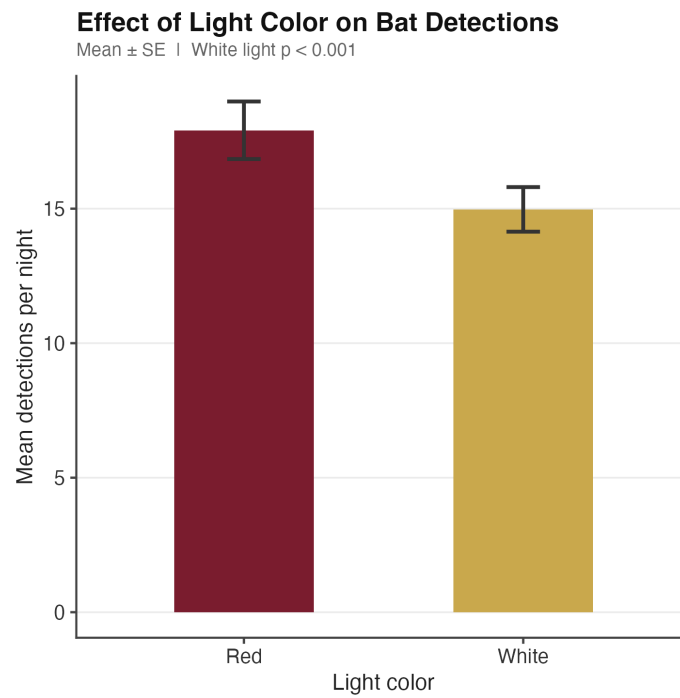


Figure 1: Bat detection counts by light color.

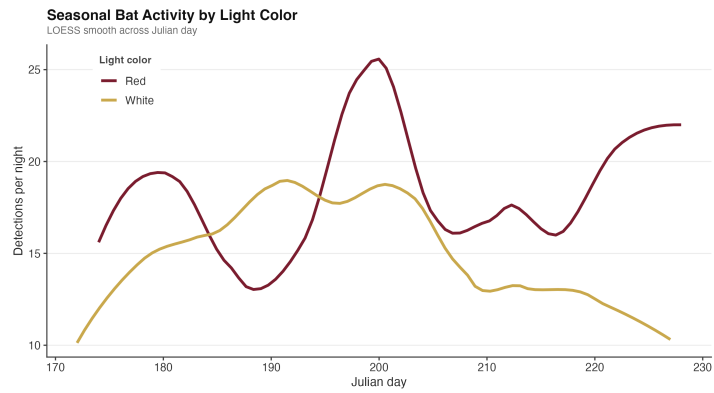


Figure 2: Bat detection counts by Julian day and light color.

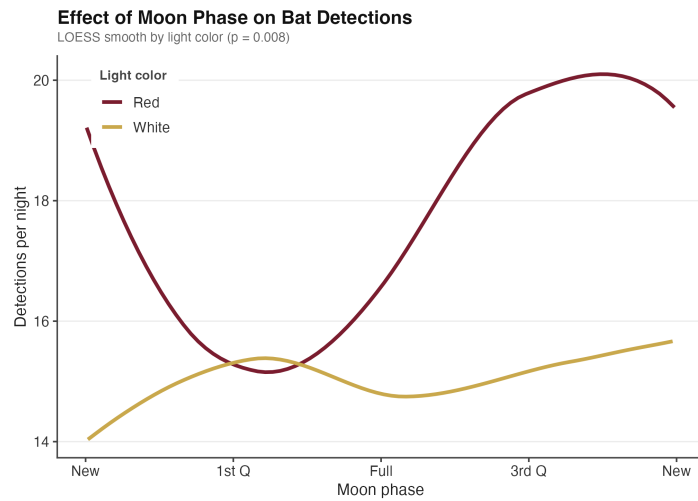


Figure 3: Bat detection counts by moon phase and light color.

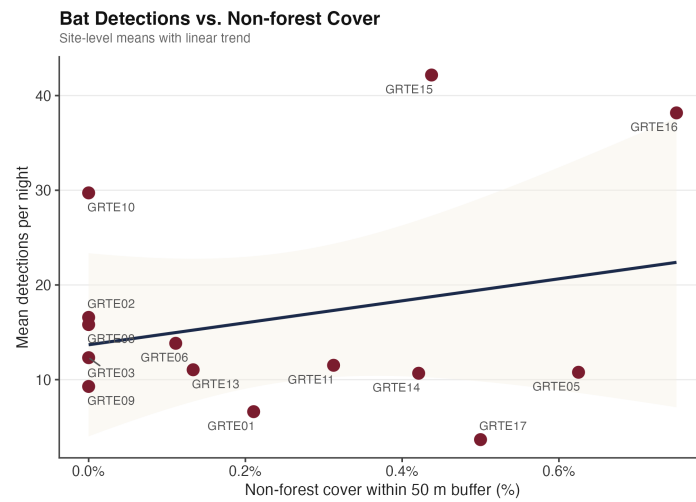


Figure 4: Bat detection counts by percent non-forest cover (percent openness) for each site.

POST-HOC: COLOR X INTENSITY

Estimated marginal means on the response scale (detections/night), averaged over distance. Holm-corrected pairwise contrasts.

=== color | intensity (response scale) ===

intensity = 10:

color	response	SE	df	asympt.LCL	asympt.UCL
R	12.7	2.29	Inf	8.93	18.1
W	13.4	2.42	Inf	9.37	19.0

intensity = 30:

color	response	SE	df	asympt.LCL	asympt.UCL
R	23.2	4.64	Inf	15.70	34.3
W	11.3	2.31	Inf	7.57	16.9

intensity = 50:

color	response	SE	df	asympt.LCL	asympt.UCL
R	18.6	3.90	Inf	12.30	28.0
W	11.4	2.45	Inf	7.49	17.4

intensity = 70:

	color	response	SE	df	asympt.LCL	asympt.UCL
R		17.8	3.21	Inf	12.49	25.4
W		16.0	2.97	Inf	11.14	23.0

intensity = 100:

	color	response	SE	df	asympt.LCL	asympt.UCL
R		13.1	2.48	Inf	9.05	19.0
W		14.0	2.51	Inf	9.87	19.9

=== Red vs. White contrast at each intensity (Holm-corrected) ===

intensity = 10:

	contrast	ratio	SE	df	null	z.ratio	p.value
R / W	0.951	0.0885	Inf	1	-0.536	0.5917	

intensity = 30:

	contrast	ratio	SE	df	null	z.ratio	p.value
R / W	2.055	0.2990	Inf	1	4.946	<0.0001	

intensity = 50:

	contrast	ratio	SE	df	null	z.ratio	p.value
R / W	1.626	0.1820	Inf	1	4.345	<0.0001	

intensity = 70:

	contrast	ratio	SE	df	null	z.ratio	p.value
R / W	1.111	0.1090	Inf	1	1.067	0.2860	

intensity = 100:

	contrast	ratio	SE	df	null	z.ratio	p.value
R / W	0.935	0.0945	Inf	1	-0.665	0.5062	

Ratio = R / W (ratio > 1 = more detections under red)

At 30% intensity, red light produced 2.06x more detections than white ($p < 0.0001$).

At 50%, 1.63x ($p = 0.002$). At 70% and 100%, no significant difference.

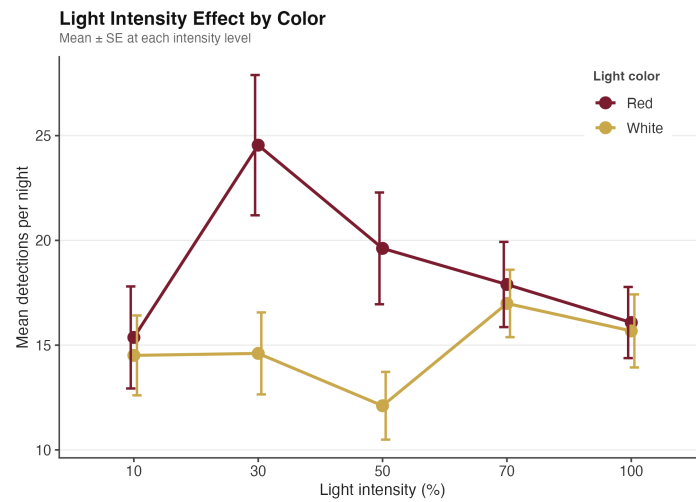


Figure 5: Bat detection counts by light intensity and light color.

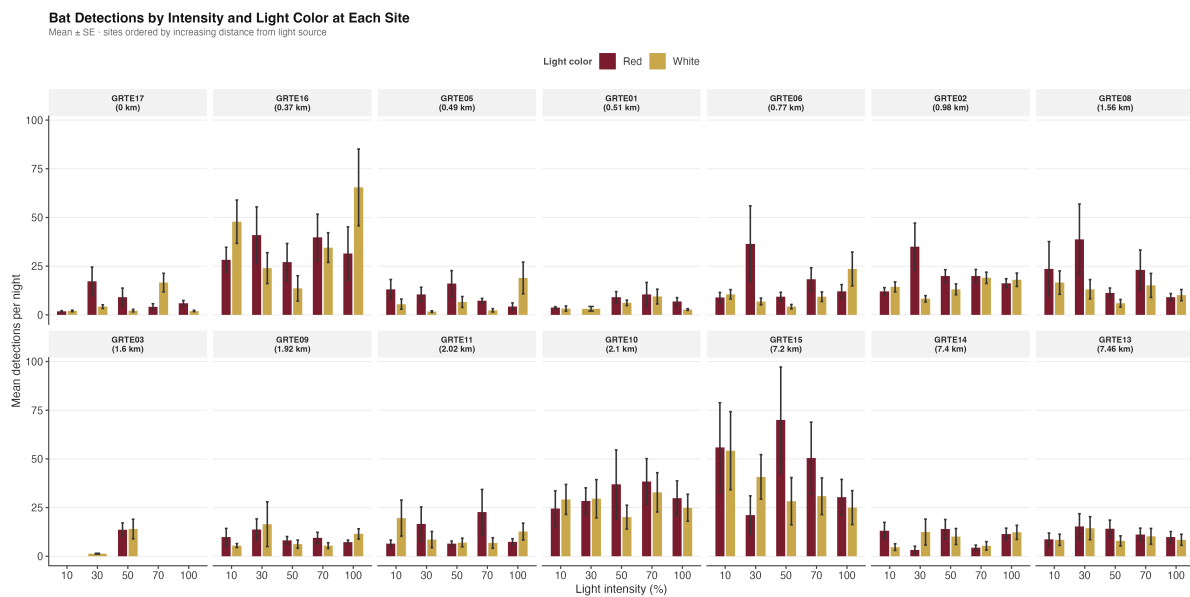


Figure 6: Bat detection counts by light intensity and light color for each site.

POST-HOC: INTENSITY WITHIN COLOR

=== Intensity pairwise contrasts within Red (Holm-corrected) ===

contrast	ratio	SE	df	z.ratio	p.value
----------	-------	----	----	---------	---------

30 / 10	1.815	0.248	Inf	4.365	<.0001	***
50 / 10	1.427	0.204	Inf	2.480	0.0590	
70 / 10	1.425	0.133	Inf	3.794	0.0011	**
100 / 10	0.999	0.116	Inf	-0.002	1.0000	
50 / 30	0.786	0.107	Inf	-1.788	0.3700	
70 / 30	0.785	0.107	Inf	-1.770	0.3700	
100 / 30	0.550	0.076	Inf	-4.362	<.0001	***
70 / 50	0.999	0.096	Inf	-0.012	1.0000	
100 / 50	0.700	0.101	Inf	-2.480	0.0590	
100 / 70	0.701	0.081	Inf	-3.026	0.0148	*

=== Intensity pairwise contrasts within White (Holm-corrected) ===

contrast	ratio	SE	df	z.ratio	p.value
30 / 10	0.880	0.081	Inf	-1.396	1.0000
50 / 10	0.873	0.080	Inf	-1.467	1.0000
70 / 10	1.260	0.117	Inf	2.495	0.1266
100 / 10	1.097	0.101	Inf	1.006	1.0000

[all remaining contrasts: p >= 0.18]

Red peaks at 30% (1.82x above 10%; $p < 0.0001$) and collapses at 100% (0.55x relative to 30%; $p < 0.0001$). White is flat across all intensities (no pairwise contrast survives Holm correction).

=== Intensity pairwise contrasts within Red (Holm-corrected) ===

color = R:

contrast	ratio	SE	df	null	z.ratio	p.value
intensity10 / intensity30	0.547	0.0749	Inf	1	-4.406	0.0001
intensity10 / intensity50	0.685	0.0977	Inf	1	-2.654	0.0478
intensity10 / intensity70	0.714	0.0664	Inf	1	-3.619	0.0024
intensity10 / intensity100	0.969	0.1130	Inf	1	-0.272	1.0000
intensity30 / intensity50	1.251	0.1920	Inf	1	1.457	0.4350
intensity30 / intensity70	1.305	0.1820	Inf	1	1.912	0.2237
intensity30 / intensity100	1.771	0.2310	Inf	1	4.387	0.0001
intensity50 / intensity70	1.043	0.1480	Inf	1	0.298	1.0000
intensity50 / intensity100	1.415	0.1710	Inf	1	2.880	0.0278
intensity70 / intensity100	1.356	0.1630	Inf	1	2.539	0.0556

color = W:

contrast	ratio	SE	df	null	z.ratio	p.value
----------	-------	----	----	------	---------	---------

intensity10 / intensity30	1.182	0.1770	Inf	1	1.119	1.0000
intensity10 / intensity50	1.170	0.1840	Inf	1	0.999	1.0000
intensity10 / intensity70	0.834	0.0851	Inf	1	-1.778	0.6033
intensity10 / intensity100	0.952	0.0962	Inf	1	-0.484	1.0000
intensity30 / intensity50	0.990	0.1350	Inf	1	-0.076	1.0000
intensity30 / intensity70	0.706	0.1000	Inf	1	-2.457	0.1402
intensity30 / intensity100	0.806	0.0989	Inf	1	-1.761	0.6033
intensity50 / intensity70	0.713	0.1040	Inf	1	-2.325	0.1809
intensity50 / intensity100	0.814	0.1090	Inf	1	-1.532	0.7535
intensity70 / intensity100	1.142	0.1170	Inf	1	1.290	0.9859

For red light, intensity 30 and 50 show increases in bat detections compared to all other intensities. We don't see intensity changing detections with white light.

POST-HOC: DISTANCE

=== Distance slope (km) by intensity level ===

intensity	dist_km.trend	SE	df	asympt.LCL	asympt.UCL
10	0.1376	0.0675	Inf	0.00538	0.270
30	0.0875	0.0689	Inf	-0.04749	0.222
50	0.1012	0.0676	Inf	-0.03139	0.234
70	0.0100	0.0670	Inf	-0.12132	0.141
100	0.0479	0.0669	Inf	-0.08316	0.179

Results are averaged over the levels of: color

Confidence level used: 0.95

=== Pairwise intensity distance-slope contrasts (Holm-corrected) ===

contrast	estimate	SE	df	z.ratio	p.value
intensity10 - intensity30	0.0501	0.0319	Inf	1.572	0.5795
intensity10 - intensity50	0.0364	0.0294	Inf	1.239	0.5965
intensity10 - intensity70	0.1275	0.0272	Inf	4.687	<0.0001
intensity10 - intensity100	0.0897	0.0269	Inf	3.335	0.0077
intensity30 - intensity50	-0.0137	0.0322	Inf	-0.425	0.6712
intensity30 - intensity70	0.0774	0.0309	Inf	2.506	0.0854
intensity30 - intensity100	0.0396	0.0304	Inf	1.301	0.5965
intensity50 - intensity70	0.0911	0.0283	Inf	3.222	0.0102
intensity50 - intensity100	0.0533	0.0276	Inf	1.931	0.3207

intensity70 - intensity100 -0.0378 0.0262 Inf -1.443 0.5965

Results are averaged over the levels of: color

P value adjustment: holm method for 10 tests

=== Intensity contrasts at close (0.3 km), medium (0.7 km), far (2.0 km) ===

dist_km = 0.3:

contrast	ratio	SE	df	null	z.ratio	p.value
intensity10 / intensity30	0.730	0.0933	Inf	1	-2.466	0.1044
intensity10 / intensity50	0.834	0.1170	Inf	1	-1.297	0.7792
intensity10 / intensity70	0.602	0.0505	Inf	1	-6.043	<0.0001
intensity10 / intensity100	0.807	0.0783	Inf	1	-2.212	0.1617
intensity30 / intensity50	1.143	0.1580	Inf	1	0.965	1.0000
intensity30 / intensity70	0.825	0.1060	Inf	1	-1.492	0.6784
intensity30 / intensity100	1.106	0.1220	Inf	1	0.913	1.0000
intensity50 / intensity70	0.722	0.0947	Inf	1	-2.483	0.1044
intensity50 / intensity100	0.968	0.1170	Inf	1	-0.274	1.0000
intensity70 / intensity100	1.340	0.1330	Inf	1	2.944	0.0292

dist_km = 0.7:

contrast	ratio	SE	df	null	z.ratio	p.value
intensity10 / intensity30	0.744	0.0912	Inf	1	-2.410	0.1275
intensity10 / intensity50	0.846	0.1150	Inf	1	-1.226	0.9528
intensity10 / intensity70	0.634	0.0497	Inf	1	-5.822	<0.0001
intensity10 / intensity100	0.836	0.0773	Inf	1	-1.936	0.3175
intensity30 / intensity50	1.137	0.1490	Inf	1	0.977	0.9528
intensity30 / intensity70	0.851	0.1050	Inf	1	-1.309	0.9528
intensity30 / intensity100	1.123	0.1160	Inf	1	1.129	0.9528
intensity50 / intensity70	0.749	0.0955	Inf	1	-2.265	0.1644
intensity50 / intensity100	0.988	0.1140	Inf	1	-0.101	0.9528
intensity70 / intensity100	1.320	0.1250	Inf	1	2.938	0.0297

dist_km = 2:

contrast	ratio	SE	df	null	z.ratio	p.value
intensity10 / intensity30	0.794	0.0902	Inf	1	-2.026	0.3418
intensity10 / intensity50	0.887	0.1160	Inf	1	-0.916	1.0000
intensity10 / intensity70	0.748	0.0520	Inf	1	-4.176	0.0003
intensity10 / intensity100	0.940	0.0801	Inf	1	-0.731	1.0000
intensity30 / intensity50	1.117	0.1280	Inf	1	0.965	1.0000

intensity30 / intensity70	0.942	0.1060	Inf	1	-0.537	1.0000
intensity30 / intensity100	1.183	0.1040	Inf	1	1.907	0.3959
intensity50 / intensity70	0.843	0.1040	Inf	1	-1.389	0.9894
intensity50 / intensity100	1.059	0.1100	Inf	1	0.556	1.0000
intensity70 / intensity100	1.256	0.1070	Inf	1	2.669	0.0685

At intensity 70, the distance gradient disappears.

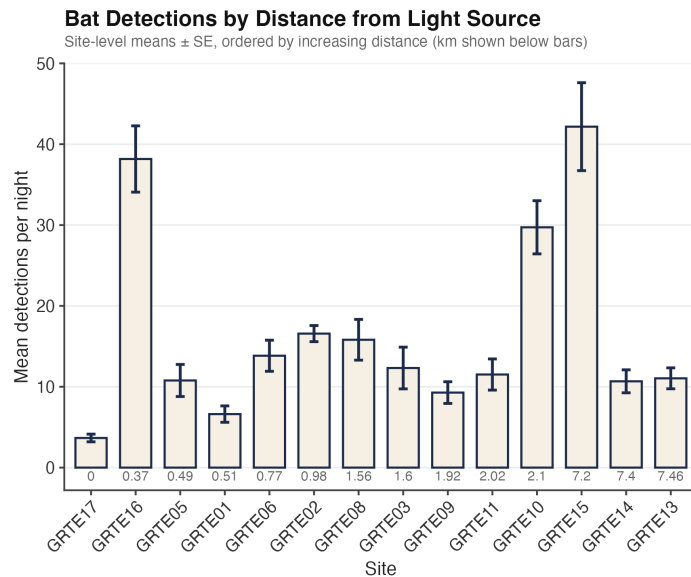


Figure 7: Bat detection counts by site and distance from Colter Bay parking lot.

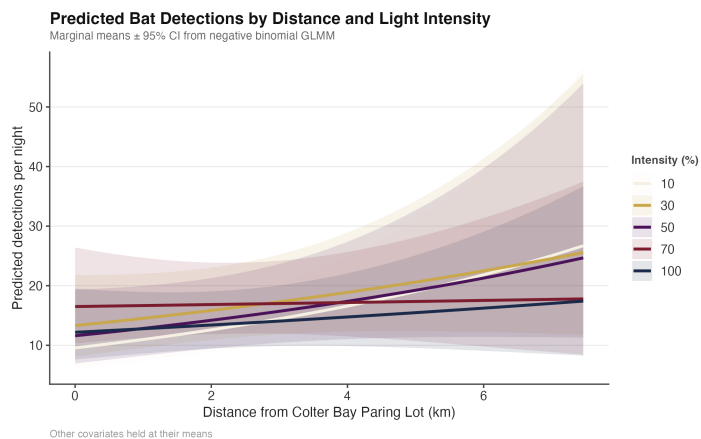


Figure 8: Bat detection counts by site, light intensity, and distance from Colter Bay parking lot.

COMMUNITY ANALYSIS

=== NMDS ===

Stress: 0.207 (k = 2, Bray-Curtis, trymax = 100)

=== PERMANOVA (adonis2) ===

adonis2(formula = perm_formula, data = env, permutations = 9999, method = "bray", strata = if (multi_ye

	Df	SumOfSqs	R2	F	Pr(>F)
Model	9	3.8338	0.15608	2.507	2e-04 ***
Residual	122	20.7295	0.84392		
Total	131	24.5633	1.00000		

Response: Distances

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Groups	1	0.00014	0.000145	0.0061	0.9378
Residuals	130	3.07863	0.023682		

Permutation test for dbrda under reduced model
Permutation: free
Number of permutations: 9999

=== dbRDA marginal tests ===

Model: dbrda(formula = comm ~ color + intensity + dist_km + mean_phase + pct_nonforest + brightness_dar

	Df	SumOfSqs	F	Pr(>F)
Model	9	3.8338	2.507	1e-04 ***
Residual	122	20.7295		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Permutation test for dbrda under reduced model
Marginal effects of terms
Permutation: free
Number of permutations: 9999

Model: dbrda(formula = comm ~ color + intensity + dist_km + mean_phase + pct_nonforest + brightness_dar

	Df	SumOfSqs	F	Pr(>F)
color	1	0.2421	1.4249	0.2008
intensity	4	1.1300	1.6626	0.0465 *
dist_km	1	1.2271	7.2217	0.0003 ***

```

mean_phase      1    0.2509 1.4769 0.1913
pct_nonforest   1    0.6241 3.6730 0.0087 **
brightness_dark 1    0.5661 3.3319 0.0123 *
Residual       122   20.7295

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Analysis of Deviance Table

=== mvabund - multivariate GLM (nBoot = 9999) ===

Model: Y ~ color + intensity + dist_km + mean_phase + pct_nonforest + brightness_dark

Multivariate test:

	Res.Df	Df.diff	Dev	Pr(>Dev)
(Intercept)	131			
color	130	1	9.22	0.414
intensity	126	4	83.52	0.002 **
dist_km	125	1	142.57	<2e-16 ***
mean_phase	124	1	24.59	0.017 *
pct_nonforest	123	1	40.16	0.001 ***
brightness_dark	122	1	54.29	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

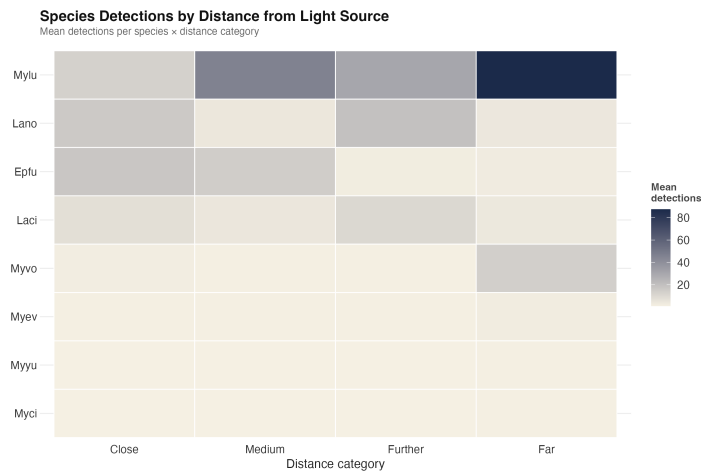


Figure 9: Bat detection counts by distance from Colter Bay parking lot.

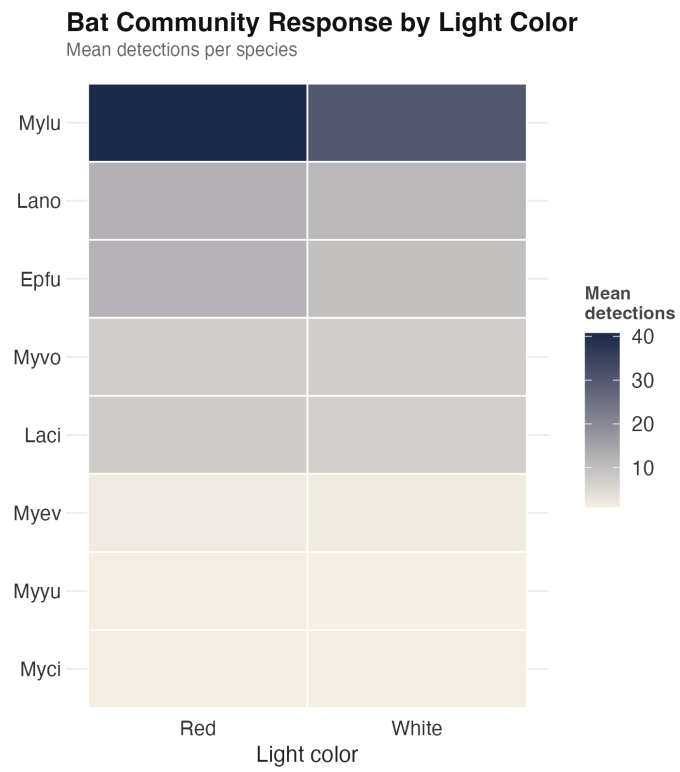


Figure 10: Bat detection counts for each species in relation to light color.

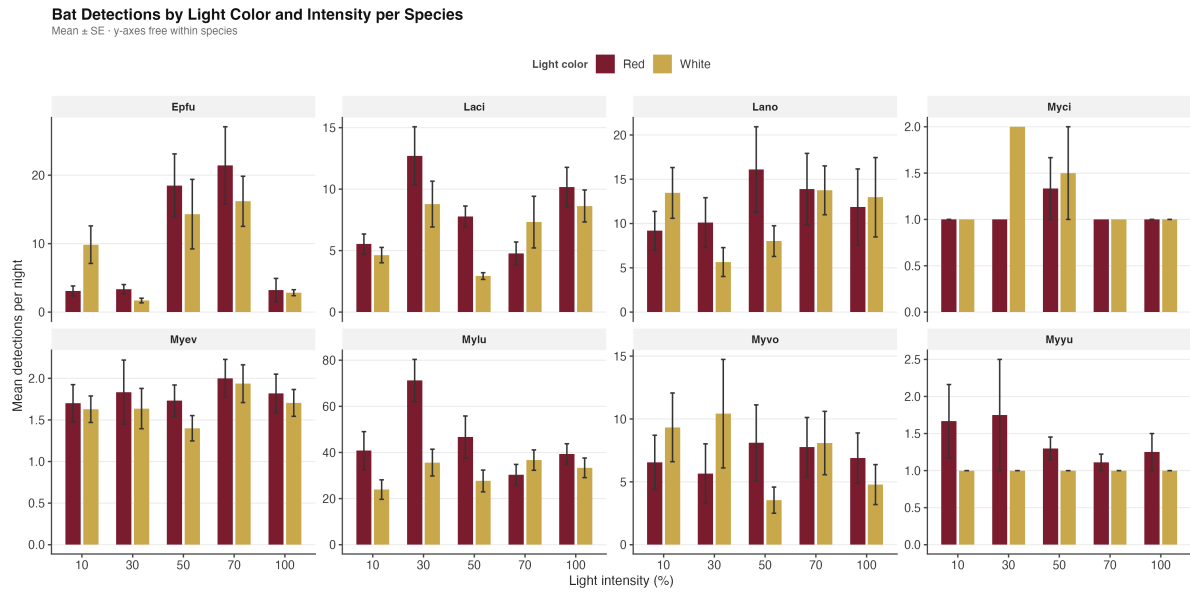


Figure 11: Bat detection counts for each species in relation to light color and light intensity.

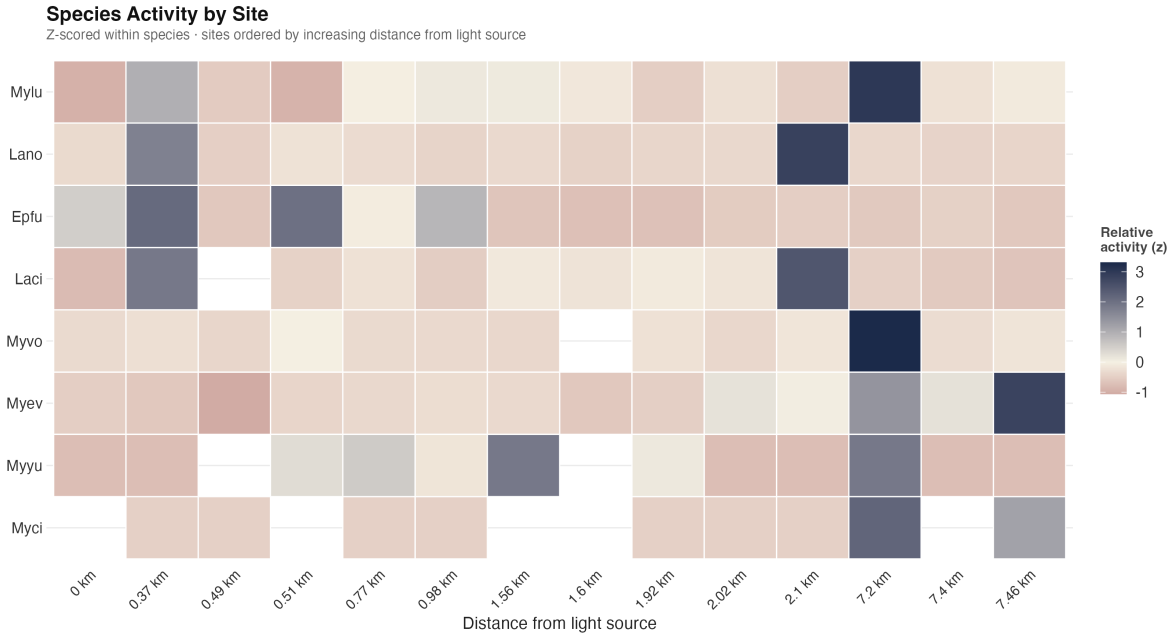


Figure 12: Bat detection counts for each species in relation to distance from Colter Bay parking lot.

Distance from Colter Bay parking lot is the strongest driver of community composition. Light color doesn't seem to impact community composition. The sky brightness at site seems important as well for community composition.

EPFU is driving the intensity effect. Myvo is driving the sight brightness effect. Distance from Colter Bay parking lot impacts each species. Myev an Myvo drive the open habitat effect. Epfu drives the moon phase effect.

SPECIES-LEVEL RESULTS

=== mvabund univariate p-values (adjusted) ===

Species	Colour	Intensity	Distance	%nonforest	brightness
Epfu	ns	***	***	ns	ns
Laci	ns	.	**	ns	ns
Lano	ns	ns	**	ns	ns
Myev	ns	ns	***	.	ns
Mylu	ns	ns	**	***	ns
Myvo	ns	ns	***	ns	***
Myyu	ns	ns	ns	ns	ns
Myci	ns	ns	ns	ns	ns

Epfu: most treatment-sensitive; responds to both intensity and distance. Mylu: tracks habitat (pct_nonforest). Myvo: tracks sky darkness. Myyu, Myci: no significant response to any measured predictor.

SUMMARY

Bat activity was significantly higher under red light than white at intermediate intensities (30%: 2.1× more detections, $p < 0.0001$; 50%: 1.6×, $p < 0.0001$), but the two light colors produced equal activity at low (10%) and high (100%) intensity, indicating that the effect of light color is intensity-dependent rather than consistent across the full brightness range. Red light showed a strong non-linear intensity response, peaking at 30% and collapsing at 100%, while white light produced flat activity across all intensities. Distance from the Colter Bay parking lot had a positive effect on bat activity overall. Bats were more active farther from the Colter Bay parking lot overall, but this distance gradient disappeared at 70% intensity. At the community level, distance was the dominant driver of species composition (dbRDA

F=7.22, p=0.0003), while light color had no detectable effect on which species were present; the intensity effect on community composition was largely attributable to Epfu, which was the most treatment-sensitive species and responded significantly to both intensity and distance, while Myvo was uniquely sensitive to ambient sky brightness and showed the strongest distance response of any species.

SPECIES KEY

Epfu	<i>Eptesicus fuscus</i>	(big brown bat)
Laci	<i>Lasiurus cinereus</i>	(hoary bat)
Lano	<i>Lasionycteris noctivagans</i>	(silver-haired bat)
Myci	<i>Myotis ciliolabrum</i>	(western small-footed myotis)
Myev	<i>Myotis evotis</i>	(long-eared myotis)
Mylu	<i>Myotis lucifugus</i>	(little brown myotis)
Myvo	<i>Myotis volans</i>	(long-legged myotis)
Myyu	<i>Myotis yumanensis</i>	(Yuma myotis)